# AutoHealOps

## Automated Monitoring & Self-Healing System

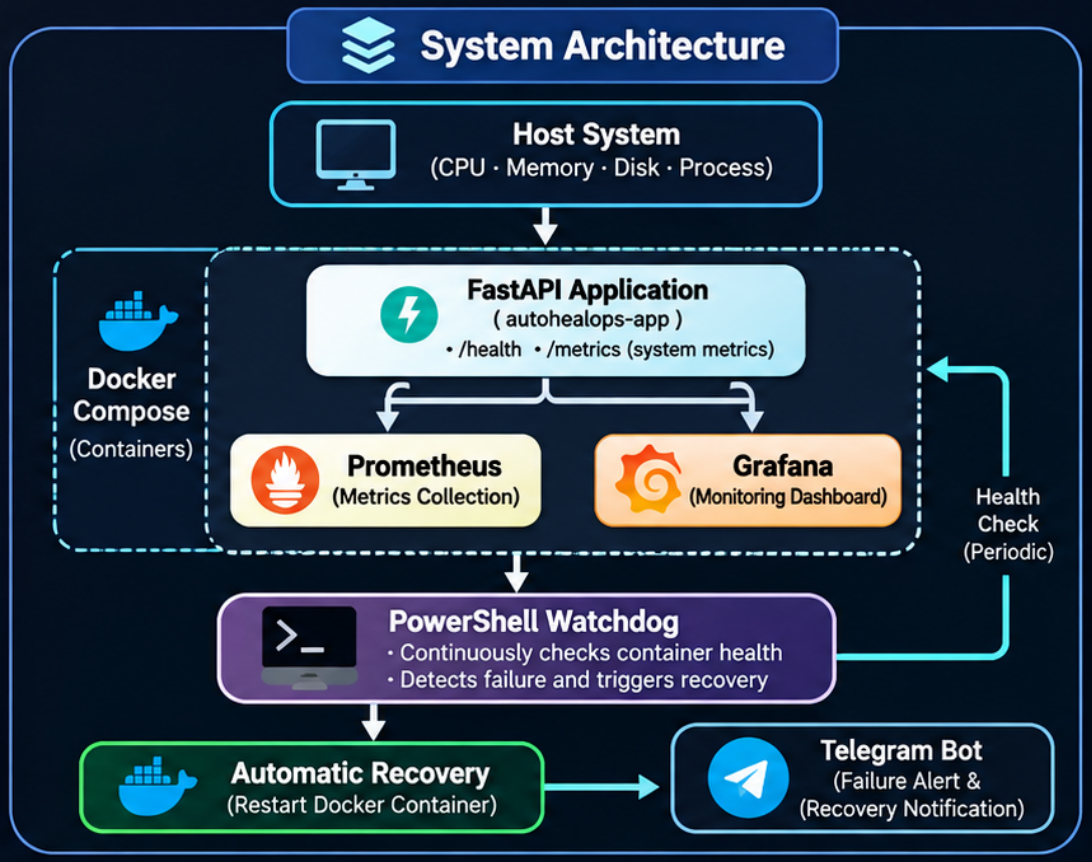
### Project Overview

A DevOps automation project that monitors application health, detects failures, automatically recovers the application, and sends real-time alerts via Telegram.

### Key Features

- Real-time system monitoring (CPU, Memory, Disk)
- Application health check (/health)
- Automatic container restart (self-healing)
- Alerts & recovery notifications via Telegram
- Containerized setup with Docker Compose
- Monitoring dashboards using Grafana

### System Architecture




### Workflow (Self-Healing Flow)

1. Application Running (Healthy)
2. Container Failure (e.g., docker stop)
3. Watchdog Detects Failure
4. Automatically Restarts Container
5. Sends Notifications (Failure & Recovery)
6. Application Restored (Healthy)

### Tech Stack

- Python (FastAPI)
- Docker & Compose
- Prometheus (Monitoring)
- Grafana (Dashboard)
- PowerShell (Watchdog)
- Telegram Bot (Alerts)